Chapter 4, Section 2

James Lee

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1. Use (2.5.3) to show that $\mathbf{P}^n$ is simply connected.

Proof. Proceed by induction on n. Let $f : X \to \mathbf{P}^n$ be an étale covering. We may assume X is connected. Then X is smooth over k since f is étale (III, 10.1), and X is projective over k since f is finite, so X is a smooth irreducible projective variety of dimension n. Let $\mathfrak{d}$ be the linear system of X corresponding to f. By (III, Ex. 11.3), almost every member of $\mathfrak{d}$ is connected. In particular, if $H \subseteq \mathbf{P}^n$ is a hyperplane, then $f^{-1}(H) \in \mathfrak{d}$ is smooth and irreducible, and the restriction $f : f^{-1}(H) \to H$ is an étale covering of $\mathbf{P}^{n-1}$. By the inductive hypothesis, the restriction of f to $f^{-1}(H)$ is an isomorphism, hence f is an isomorphism. $\square$

2. *Classification of Curves of Genus 2.* Fix an algebraically closed field k of characteristic $\neq 2$.

(a) If X is a curve of genus 2 over k, the canonical linear system $|K|$ determines a finite morphism $f : X \to \mathbf{P}^1$ of degree 2 (Ex. 1.7). Show that it is ramified at exactly 6 points, with ramification index 2 at each one. Note that f is uniquely determined, up to an automorphism of $\mathbf{P}^1$, so X determines an (unordered) set of 6 points of $\mathbf{P}^1$, up to an automorphism of $\mathbf{P}^1$.

(b) Conversely, given six distinct elements $\alpha_1, \ldots, \alpha_6 \in k$, let K be the extension of $k(x)$ determined by the equation $z^2 = (x - \alpha_1) \cdots (x - \alpha_6)$. Let $f : X \to \mathbf{P}^1$ be the corresponding morphism of curves. Show that $g(X) = 2$, the map f is the same as the one determined by the canonical linear system, and f is ramified over the six points $x = \alpha_i$ of $\mathbf{P}^1$, and nowhere else.

(c) Using (I, Ex. 6.6), show that if P_1, P_2, P_3 are three distinct points of $\mathbf{P}^1$, then there exists a unique $\varphi \in \operatorname{Aut} \mathbf{P}^1$ such that $\varphi(P_1) = 0$, $\varphi(P_2) = 1$, $\varphi(P_3) = \infty$. Thus in (a), if we order the six points of $\mathbf{P}^1$, and then normalize by sending the first three to $0, 1, \infty$, respectively, we may assume that X is ramified over $0, 1, \infty, \beta_1, \beta_2, \beta_3$, where $\beta_1, \beta_2, \beta_3$ are three distinct elements of k, $\neq 0, 1$.

(d) Let Σ_6 be the symmetric group on 6 letters. Define an action of Σ_6 on sets of three distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, as follows: reorder the set $0, 1, \infty, \beta_1, \beta_2, \beta_3$ according to a given element $\sigma \in \Sigma_6$, then renormalize as in (c) so that the three become $0, 1, \infty$ again. Then the last three are the new $\beta_1', \beta_2', \beta_3'$.

(e) Summing up, conclude that there is a one-to-one correspondence between the set of isomorphism classes of curves of genus 2 over k, and triples of distinct elements $\beta_1, \beta_2, \beta_3$ of k, $\neq 0, 1$, modulo the action of Σ_6 described in (d). In particular, there are many non-isomorphic curves of genus 2. We say that curves of genus 2 depend on three parameters, since they correspond to the points of an open subset of $\mathbf{A}_k^3$ modulo a finite group.

Proof.

(a) By Hurtwitz's theorem, the degree of the ramification divisor is computed to be

$$\deg R = 2g(X) - 2 - 2(2g(\mathbf{P}^1) - 2) = 6.$$

The ramification index at a point cannot exceed the degree of the morphism, so f must be ramified at exactly 6 distinct points each having ramification index 2.

(b) Let $B = k[x, z]/(z^2 - g(x))$ be an affine coordinate ring of X with $g(x) = (x - \alpha_1) \cdots (x - \alpha_6)$, and let f be the projection morphism onto the x-axis. By (II, 8.3A), we have

$$\Omega_{B/k} \cong (B\mathrm{d}x \oplus B\mathrm{d}z)/(g'(x)\mathrm{d}x + 2z\mathrm{d}z).$$

Let $P = (x', z')$ be a closed point in X. If $z' \neq 0$, then z is invertible in $\mathcal{O}_{P,X}$, so that

$$\mathrm{d}z = -\frac{g'(x)}{2z}\mathrm{d}x,$$

which means $\mathrm{d}x$ generates $(\Omega_{B/k})_P$. If $z' = 0$, then $x' = \alpha_i$ for some i, and $g'(x)$ is invertible in $\mathcal{O}_{P,X}$, so that

$$\mathrm{d}x = -\frac{2z}{g'(x)}\mathrm{d}z,$$

which means $\mathrm{d}z$ generates $(\Omega_{B/k})_P$. Since $\mathrm{d}x$ is a local parameter of $\mathbf{P}^1$, we see that P is ramified if and only if $z' = 0$ with ramification index $e_P = v_P(\mathrm{d}x/\mathrm{d}z) + 1 = 2$. There are six ramification points corresponding to the six roots of $g(x)$. Hence, $g(X) = 2$ by the Hurwitz formula.

□

3. *Plane Curves.* Let X be a curve of degree d in $\mathbf{P}^2$. For each $P \in X$, let $T_P(X)$ be the tangent line to X at P (I, Ex.7.3). Considering $T_P(X)$ as a point of the dual projective plane $(\mathbf{P}^2)^*$, the map $P \to T_P(X)$ gives a morphism of X to its *dual curve* X^* in $(\mathbf{P}^2)^*$ (I, Ex. 7.3). Note that even though X is nonsingular, X^* in general will have singularities. We assume $\operatorname{char} k = 0$ below.

(a) Fix a line $L \subseteq \mathbf{P}^2$ which is not tangent to X. Define a morphism $\varphi : X \to L$ by $\varphi(P) = L \cap T_P(X)$, for each point $P \in X$. Show that φ is ramified at P if and only if either (1) $P \in L$, or (2) P is an *inflection point* of X, which means that the intersection multiplicity (I, Ex. 5.4) of $T_P(X)$ with X at P is ≥ 3. Conclude that X has only finitely many inflection points.

(b) A line of $\mathbf{P}^2$ is a *multiple tangent* of X if it is tangent to X at more than one point. It is a *bitangent* if it is tangent to X at exactly two points. If L is a multiple tangent of X, tangent to X at the points $P_1, \ldots, P_r$, and if none of the P_i is an inflection point, show that the corresponding point of the dual curve X^* is an *ordinary r-fold point*, which means a point of multiplicity r with distinct tangent directions (I, Ex. 5.3). Conclude that X has only finitely many multiple tangents.

(c) Let $O \in \mathbf{P}^2$ be a point which is not on X, nor on any inflectional or multiple tangent of X. Let L be a line not containing O. Let $\psi : X \to L$ be the morphism defined by projection from O. Show that ψ is ramified at $P \in X$ if and only if the line OP is tangent to X at P, and that the ramification index is 2. Use Hurwitz's theorem and (I, Ex. 7.2) to conclude that there are exactly $d(d-1)$ tangents of X passing through O. Hence, the degree of the dual curve (sometimes called the *class* of X) is $d(d-1)$.

(d) Show that for all but a finite number of points of X, a point O of X lies on exactly $(d+1)(d-2)$ tangents of X, not counting the tangent at O.

(e) Show that the degree of the morphism φ of (a) is $d(d-1)$. Conclude that if $d \geq 2$, then X has $3d(d-2)$ inflection points, properly counted. (If $T_P(X)$ has intersection multiplicity r with X at P, then P should be counted $r-2$ times as an inflection point. If $r = 3$, we called it an *ordinary inflection point.*) Show that an ordinary inflection point of X corresponds to an ordinary cusp of the dual curve X^*.

(f) Now let X be a plane curve of degree $d \geq 2$, and assume that the dual curve X^* has only nodes and ordinary cusps as singularities (which should be true for sufficiently general X.) Then show that X has exactly $\frac{1}{2}d(d-2)(d-3)(d+3)$ bitangents.

(g) For example, a plane cubic curve has exactly 9 inflection points, all ordinary. The line joining any two of the intersects the curve in a third one.

(h) A plane quartic curve has exactly 28 bitangents. (This holds even if the curve has a tangent with four-fold contact, in which case the dual curve X^* has a tacnode.)

Proof.

(a) Find local coordinates x, y of $\mathbf{P}^2$ such that X is given by $y = f(x)$, L is defined by $y = 0$, and P is not mapped to the point of infinity of L. Setting $u = x$ and $t = x$ to be local parameters for X and L, respectively, φ and $\varphi^*\mathrm{d}t$ can locally be written in the form

$$\varphi(u) = u - f/f', \quad \varphi^*\mathrm{d}t = \frac{\mathrm{d}t}{\mathrm{d}u}\mathrm{d}u = \frac{ff''}{(f')^2}\mathrm{d}t.$$

By (2.2), φ is ramified at P if and only if $v_P(\mathrm{d}t/\mathrm{d}u) > 0$. Since v_P is multiplicative, we have

$$v_P\left(\frac{\mathrm{d}t}{\mathrm{d}u}\right) = v_P(f) + v_P(f'') - 2v_P(f').$$

By definition, f is zero at P if and only if $P \in L$, and 1 otherwise. Also, $\varphi(P)$ is the point of infinity of L if and only if $f'(P) = 0$, which implies $v_P(f') = 0$. The last assertion to prove is $v_P(f'') > 0$ if and only if P is an inflection point of X. We prove that the intersection multiplicity $\mu_P(X, T_P(X))$ and the ordering of vanishing $v_P(f'')$ is related in the following way

$$v_P(f'') + 2 = \mu_P(X, T_P(X)).$$

Suppose $P = (0, f(0))$, and recall the following

$$X : y - f(x) = 0, \quad T_P(X) : y - (f'(0)x + f(0)) = 0.$$

By definition, we have

$$\begin{aligned} \mu_p(X, T_P(X)) &= \operatorname{length}\left(k[x]/(f(x) - (f'(0)x + f(0)))\right)_{(x)} \\ &= \text{largest } d > 0 \text{ such that } x^d \mid f(x) - (f'(0)x + f(0)) \\ &= v_P(f(x) - (f'(0)x + f(0))) \\ &= v_P(x^2 f'') \\ &= 2 + v_P(f''). \end{aligned}$$

Hence, $v_P(\mathrm{d}t/\mathrm{d}u) > 0$ if and only if $P \in L$ or P is an inflection point of X. Since $\varphi : X \to L$ ramifies at finitely many points because $\operatorname{char} k = 0$, and φ ramifies at every inflection point, we conclude that X has only finitely many inflection points.

(b) Find local coordinates x, y such that L is defined by $y = 0$, and X intersects L at non-inflection points $P_1, \ldots, P_r$, let ξ, η be dual coordinates of $(\mathbf{P}^2)^*$, and let $f(x, y) = 0$ be a defining equation of X. We can locally write $\phi : X \to X^* \subseteq (\mathbf{P}^2)^*$ in terms of the partial derivatives of f

$$P \mapsto \phi(P) = (f_x(P), f_y(P)).$$

Remark 1. Notice that ϕ is given by the linear system in $\mathcal{O}_X(d-1)$ spanned by the partial derivatives of f. In particular, $\mathcal{O}_X(d-1)$ has degree $d(d-1)$ as a line bundle on X, so the dual curve $X^* = \phi(X)$ is a curve of degree $d(d-1)$ in $(\mathbf{P}^2)^*$.

The image of the tangent vector of X at P_i can be computed via the Hessian of f, which is given by

$$v_i = \begin{pmatrix} f_{xx} \\ f_{xy} \end{pmatrix}_{P_i} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{pmatrix}_{P_i} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in T_Q(X^*),$$

and each v_i corresponds to a tangent vector of X^* at Q. We want to show each v_i is (1) non-zero and (2) distinct.

(1) By definition, P_i is an inflection point if and only if $f_{xx}(P_i) = 0$. Hence, v_i is non-zero.

(2) Without loss of generality, assume $P_1 = (0, 0)$. Suppose there exists $P_2 = (x', 0)$ such that v_1 and v_2 differ by a constant multiple. This means we have

$$f_{xx}(0,0) f_{xy}(x', 0) - f_{xx}(x', 0) f_{xy}(0, 9) = 0$$

(c) Find coordinates x, y, z such that $O = (0 : 1 : 0)$, L is the x-axis defined by $y = 0$, ψ is the projection onto x-axis, $f(x, y) = 0$ be a locally defining equation of X with a point P in X. Let u be a local parameter of X at P. We want to determine when $v_p(\mathrm{d}x/\mathrm{d}u) > 0$. We have two cases to consider:

$\boxed{\partial_x f(P) = 0}$ The tangent line is not vertical, so x is a local parameter of X at P, which means ψ cannot be ramified at P.

$\boxed{\partial_x f(P) \neq 0}$ The tangent line is not horizontal, so we can locally describe x as a function of y, i.e., y is a local parameter of X at P. Using the relation $\mathrm{d}f = 0$, we find that

$$\frac{\mathrm{d}x}{\mathrm{d}y} = -\frac{\partial_y f}{\partial_x f}\mathrm{d}y, \quad v_P\left(\frac{\mathrm{d}x}{\mathrm{d}y}\right) = v_P(\partial_y f).$$

Hence, ψ is ramified at P if and only if $v_P(\partial_y f) > 0$ if and only if $\partial_y f(P) = 0$ if and only if the tangent line of P intersects O. In this case, the intersection multiplicity of OP with X is < 2,

The degree of ψ is d, so the degree of the ramification index of ψ is $d(d-1)$ by Hurwitz's theorem. Hence, X has exactly $d(d-1)$ tangents passing through O.

(d)

(e)

(f)

(g)

(h) $x^4 + y^4 + z^4 = 0$, $x^3 y + y^3 z + z^3 x = 0$

□

5. *Automorphisms of a Curve of Genus ≥ 2.* Prove the theorem of Hurtwitz [1] that a curve X of genus $g \geq 2$ over a field of characteristic 0 has at most $84(g-1)$ automorphisms. We will see later (Ex. 5.2) or (V, Ex. 1.11) that the group $G = \operatorname{Aut} X$ is finite. So let G have order n. Then G acts on the function field $K(X)$. Let L be the fixed field. Then the field extension $L \subseteq K(X)$ corresponds to a finite morphism of curves $f : X \to Y$ degree n.

(a) If $P \in X$ is a ramification point, and $e_p = r$, show that $f^{-1}f(P)$ consists of exactly n/r points, each having ramification index r. Let $P_1, \ldots, P_s$ be a maximal set of ramification points of X lying over distinct points of Y, and let $e_{P_i} = r_i$. Then show that Hurtwitz's theorem implies that

$$(2g-2)/n = 2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i).$$

(b) Since $g \geq 2$, the left hand side of the equation is > 0. Show that if $g(Y) \geq 0$, $s \geq 0$, $r_i \geq 2$, $i = 1, \ldots, s$ are integers such that

$$2g(Y) - 2 + \sum_{i=1}^{s}(1 - 1/r_i) > 0,$$

then the minimum value of this expression is $1/42$. Conclude that $n \leq 84(g-1)$.

Proof.

(a) Each $\sigma \in G$ induces an automorphism of X such that $f \circ \sigma = f$, so there is a group action of G on each fibre $f^{-1}f(P)$. The action is transitive (A.M., Ex. 5.13), and we have $v_{P_i}(f(P_i)) = v_{\sigma(P_i)}(f(\sigma(P_i)))$ because σ induces an isomorphism of local rings $\mathcal{O}_{P_i,X} \cong \mathcal{O}_{\sigma(P_i),X}$. The equation follows from

$$\begin{aligned}
\sum_{P \in Y}(e_P - 1) &= \sum_{i=1}^{s} \sum_{Q \in f^{-1}f(P_i)} (e_Q - 1) \\
&= \sum_{i=1}^{s} \frac{n}{r_i}(r_i - 1) \\
&= n\sum_{i=1}^{s}(1 - 1/r_i).
\end{aligned}$$

(b) The following give the minimum value

$$g(Y) = 0, \quad s = 3, \quad r_1 = \frac{1}{2}, \quad r_2 = \frac{1}{3}, \quad r_3 = \frac{1}{7}.$$

We conclude that

$$n \leq 42(2g-2) = 84(g-1).$$

□

6. f_* *for Divisors.* Let $f : X \to Y$ be a finite morphism of curves of degree n. We define a homomorphism $f_* : \operatorname{Div} X \to \operatorname{Div} Y$ by $f_*(\sum n_i P_i) = \sum n_i f(P_i)$ for any divisor $D = \sum n_i P_i$ on X.

(a) For any locally free sheaf $\mathcal{E}$ on Y, of rank r, we define $\det \mathcal{E} = \wedge^r \mathcal{E} \in \operatorname{Pic} Y$ (II, Ex. 6.11). In particular, for any invertible sheaf $\mathcal{M}$ on X, $f_*\mathcal{M}$ is locally free of rank n on Y, so we can consider $\det f_*\mathcal{M} \in \operatorname{Pic} Y$. Show that for any divisor D on X,

$$\det(f_*\mathcal{O}_X(D)) \cong (\det f_*\mathcal{O}_X) \otimes \mathcal{O}_Y(f_*D).$$

Note in particular that $\det(f_*\mathcal{O}_X(D)) \neq \mathcal{O}_Y(f_*D)$ in general!.

(b) Conclude that f_*D depends only on the linear equivalence class of D, so there is an induced homomorphism $f_* : \operatorname{Pic} X \to \operatorname{Pic} Y$. Show that $f_* \circ f^* : \operatorname{Pic} Y \to \operatorname{Pic} Y$ is just multiplication by n.

(c) Use duality for a finite flat morphisms (III, Ex. 6.10) and (III, Ex. 7.2) to show that

$$\det f_*\Omega_X \cong (\det f_*\mathcal{O}_X)^{-1} \otimes \Omega_Y^{\otimes n}.$$

(d) Now assume that f is separable, so we have the ramification divisor R. We define the *branch divisor* B to be the divisor f_*R on Y. Show that

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(-B).$$

Proof.

(a) Consider the case when D is an effective divisor. The pushfoward f_* is exact for finite morphisms, so applying f_* to the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0$$

gives us another exact sequence

$$0 \longrightarrow f_*(\mathcal{O}_X(-D)) \longrightarrow f_*\mathcal{O}_X \longrightarrow f_*\mathcal{O}_D \longrightarrow 0.$$

(b) It suffices to show $f_*f^*Q = nQ$ for every point $Q \in Y$. By definition, $f_*f^*Q = n'Q$ for some $n' \geq 0$. On the otherhand, $\deg f_*D = \deg D$, and $\deg f^*D = n \deg D$ (II, 6.9), so $\deg f_*f^*D = n \deg D$. Hence, $n' = n$.

(c)

(d) Let K_X and K_Y be canonical divisors of X and Y, respectively. They satisfy the following relation (2.3)

$$f^*K_Y - K_X \sim -R.$$

Applying f_* gives

$$nK_Y - f_*K_X \sim -B,$$

where we used the fact that $f_*f^*K_Y = nK_Y$ (b). On the otherhand, the results of (a) applied to $D = K_X$ and (c) give

$$\mathcal{O}_Y(nK_Y) \cong (\det f_*\mathcal{O}_X) \otimes (\det f_*\Omega_X)$$
$$, \quad \mathcal{O}_Y(-f_*K_X) \cong (\det f_*\Omega_X) \otimes (\det f_*\Omega_X)^{-1}.$$

Combining these two isomorphisms gives

$$(\det f_*\mathcal{O}_X)^2 \cong \mathcal{O}_Y(nK_Y - f_*K_X) \cong \mathcal{O}_Y(-B).$$

□

7. *Étale Covers of Degree 2.* Let Y be a curve over a field k of characteristic $\neq 2$. We show there is a one-to-one correspondence between finite étale morphisms $f : X \to Y$ of degree 2, and 2-torsion elements of $\operatorname{Pic} Y$, i.e., invertible sheaves $\mathcal{L}$ on Y with $\mathcal{L}^2 \cong \mathcal{O}_Y$.

(a) Given an étale morphism $f : X \to Y$ of degree 2, there is a natural map $\mathcal{O}_Y \to f_*\mathcal{O}_X$. Let $\mathcal{L}$ be the cokernel. Then $\mathcal{L}$ is an invertible sheaf on Y, $\mathcal{L} \cong \det f_*\mathcal{O}_X$, and so $\mathcal{L}^2 \cong \mathcal{O}_X$ by (Ex. 2.6). Thus an étale cover of degree 2 determines a 2-torsion element in $\operatorname{Pic} Y$.

(b) Conversely, given a 2-torsion element $\mathcal{L}$ in $\operatorname{Pic} Y$, define an $\mathcal{O}_Y$-algebra structure on $\mathcal{O}_Y \oplus \mathcal{L}$ by $\langle a, b\rangle \cdot \langle a', b'\rangle = \langle aa' + \varphi(b \otimes b'), ab' + a'b\rangle$, where φ is an isomorphism of $\mathcal{L} \otimes \mathcal{L} \to \mathcal{O}_Y$. Then take $X = \mathbf{Spec}(\mathcal{O}_Y \oplus \mathcal{L})$. Show that X is an étale cover of Y.

(c) Show that these two processes are inverse to each other.

Proof.

(b) For each point $y \in Y$, we have $X_y \cong \operatorname{Spec}(\mathcal{O}_y^{\oplus 2})$ with $\mathcal{O}_y$-algebra structure given by

$$\langle a, b\rangle \cdot \langle a', b'\rangle = \langle aa' + bb', ab' + a'b\rangle.$$

In particular, the natural morphism $f : X \to Y$ is flat because $f_*\mathcal{O}_X$ is locally free of rank 2, and the structure morphism $\mathcal{O}_y \to \mathcal{O}_y^{\oplus 2}$ is given by $1 \mapsto \langle 1, 0\rangle$.

(c)

$\square$